

Water Processing & Environmental Sustainability

Engineering Chemistry (BSSH105) · Unit I — Mind Map Overview

Unit I — Four Study Areas

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graph TD; A[Unit I — Four Study Areas] --- B[Water Sources & Quality]; A --- C[Boiler Problems]; A --- D[Water Softening Methods]; A --- E[Sustainability];
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Water Sources & Quality

Sources · impurities · hardness · alkalinity · TDS · DO · COD

Boiler Problems

Scale & sludge · caustic embrittlement · priming & foaming

Water Softening Methods

Zeolite process · ion exchange · ultrafiltration

Sustainability

Environmental · three pillars · sustainable development · pollution types

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Water Sources, Impurities & Quality

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Sources of water

Ground, surface, sea and rain water

Impurities in water

Biological (bacteria), suspended (sand, clay), colloidal (oil, organic matter), dissolved (gases, Ca/Mg salts)

Hardness — the soap-consuming capacity of water

Caused by dissolved Ca/Mg minerals, CO_2 action and dissolved O_2

Measured in ppm, mg/L, °Clarke, °French (1 ppm = 1 mg/L = 0.07°Cl = 0.1°Fr)

Determined by EDTA (complexometric) titration

Drinking water limit: 150–200 ppm; boiler/pharma: 0–10 ppm

Alkalinity, TDS, DO & COD

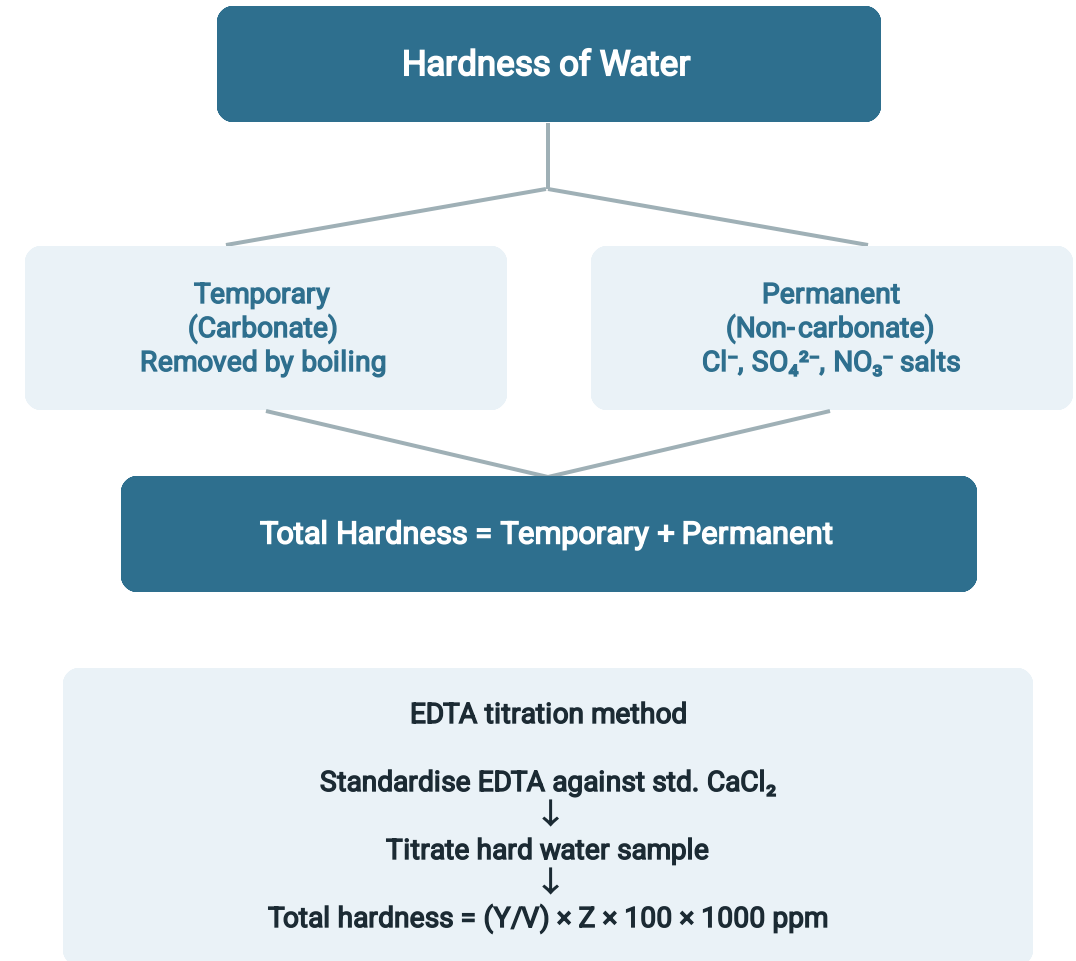
Alkalinity: found via phenolphthalein + methyl orange titration; limit ≤ 200 ppm

TDS: dissolved particles; drinking water 25–250 mg/L (max 500)

DO: oxygen available to aquatic life, in mg O_2 /L

COD: mg of O_2 to oxidise impurities in 1000 mL sample

Hardness of water — at a glance



Boiler Problems from Hard Water

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Scale & sludge

Scale: hard, adherent deposit baked onto boiler surfaces

Sludge: loose, soft deposit that can be removed easily

Caustic embrittlement

Na_2CO_3 hydrolyses to NaOH , which attacks stressed boiler joints and makes iron brittle

Prevented by using Na_3PO_4 instead of Na_2CO_3 , adding lignin/tannins or Na_2SO_4 , and keeping pH 8–9

Priming & foaming

Priming: water droplets carried with steam during rapid/sudden boiling

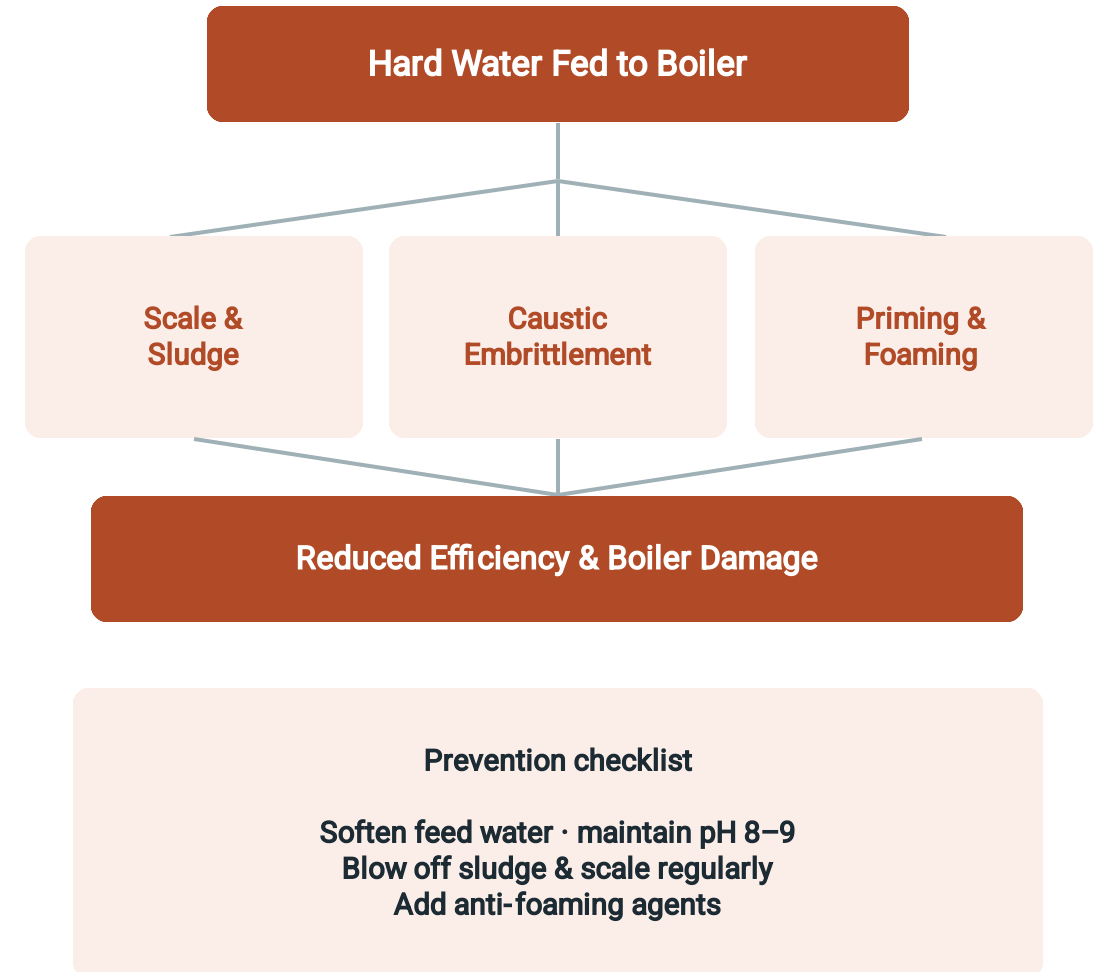
Foaming: persistent surface bubbles caused by oils & alkalies

Prevention: control steaming rate, blow off sludge/scale regularly, add anti-foaming agents (e.g. castor oil)

Consequences

Reduced turbine/heater efficiency, machinery wear, unreliable water-level readings

How hard water damages a boiler



Water Softening Methods (External Treatment)

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Zeolite / Permutit process

Sodium zeolite exchanges Na^+ for Ca^{2+} / Mg^{2+} in the water

Natural (non-porous) or synthetic (porous) zeolite

Regenerated by passing 5–10% brine (NaCl) through it

Ion exchange / demineralisation

Cation exchanger $[\text{RH}_2]$ swaps H^+ for Ca^{2+} , Mg^{2+} , Na^+ , Fe^{2+}

Anion exchanger $[\text{R}'(\text{OH})_2]$ swaps OH^- for anions; degasifier removes CO_2

Regenerated with dil. HCl (cation) and KOH (anion)

Gives very low hardness (0–2 ppm) but has high capital cost

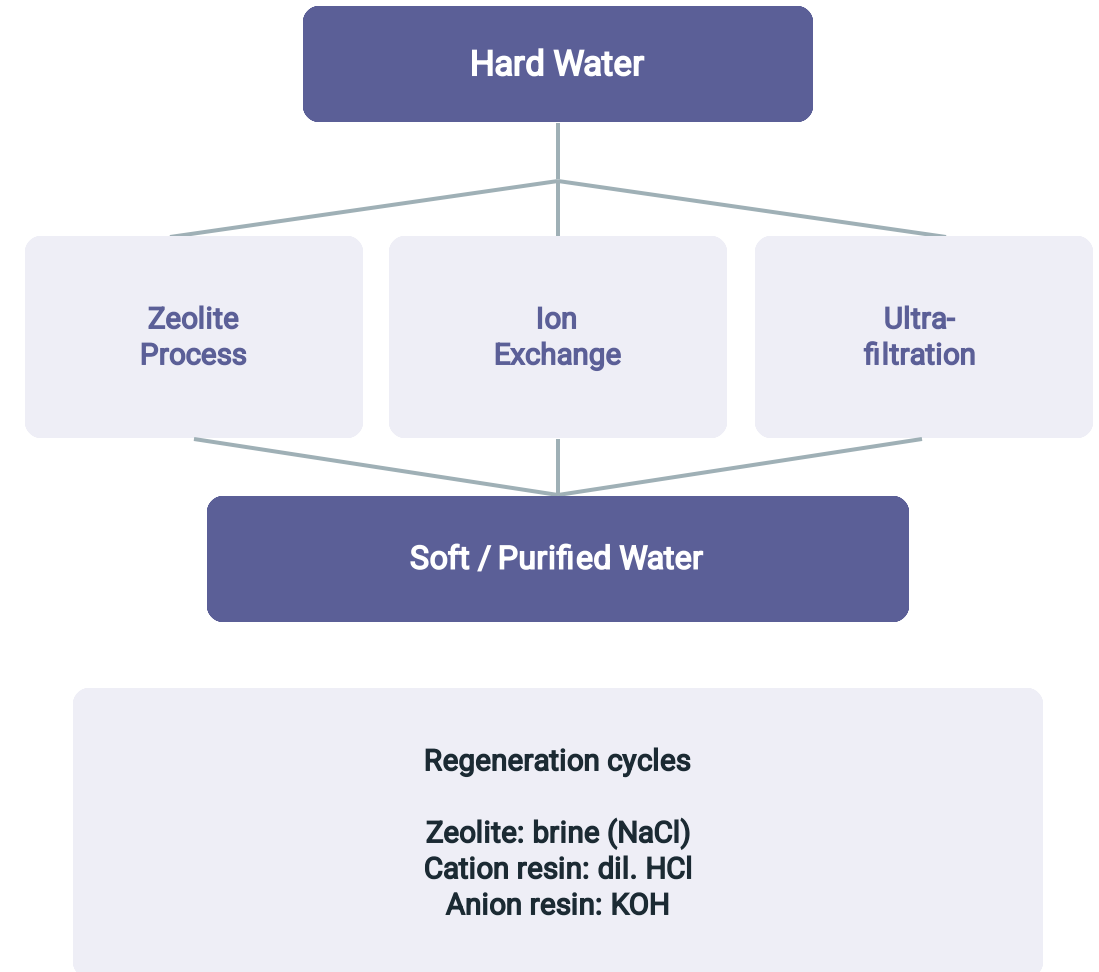
Ultrafiltration (UF)

Semi-permeable membrane + hydrostatic pressure removes suspended solids, bacteria, viruses

Pore size 0.1–0.01 micron, rated by MWCO

Zero power needed, low maintenance, 90–100% pathogen removal

Three routes to soft water



Environmental Sustainability

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What is sustainability

Meeting today's needs without compromising future generations

Rests on three pillars: environmental, economic, social

Sustainable development

Meeting human development goals while sustaining natural systems

Benefits of sustainability

Secures resources for the future, cuts energy use, benefits society, encourages reusable products and recycling

Environmental pollution

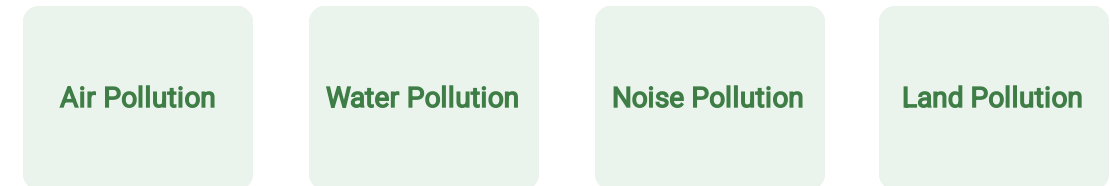
Any undesirable change harmful to life or property

Four types: air, water, noise, land pollution

The three pillars of sustainability



Types of environmental pollution



Why it matters

Protects ecosystems and natural resources
Supports long-term industrial & social wellbeing